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CS 499 Milestone Three Narrative

Artifact Description

The artifact selected for this enhancement is the Security Threat Monitoring Dashboard, a full stack security monitoring application created for my CS 499 capstone project. The application was originally developed as a backend security monitoring system using Node.js, Express, MongoDB Atlas, Mongoose, JWT authentication, and bcrypt password hashing. In the previous milestone, the artifact was improved with a frontend dashboard that allows users to authenticate, create security logs, view dashboard metrics, and review generated alerts.

For this milestone, I focused on enhancing the artifact in the category of algorithms and data structures. The purpose of this enhancement was to try to improve the way the system evaluates security events and prioritizes alerts. Instead of relying only on simple rule based severity labels, the enhanced version now uses weighted threat scoring, repeated event analysis, suspicious IP ranking, and alert prioritization. These adjustments make it more complex.

Justification for Inclusion in the ePortfolio

I selected this artifact for my ePortfolio because it aligns very closely with my career goals in cybersecurity, cloud security, and security operations. Security monitoring systems depend heavily on algorithms to analyze large amounts of event data and identify suspicious behavior. This artifact gives me the opportunity to try and demonstrate practical algorithmic thinking in a security focused type of application.

The enhanced artifact helps showcase my skills in algorithms and data structures by using weighted scoring, conditional logic, aggregation, sorting, ranking, and event correlation. The threat scoring algorithm assigns different point values to different security event types. For example, a failed login receives a lower score than an admin access attempt, while repeated failed logins from the same IP address increase the final threat score. The system then uses the score to classify events as Low, Medium, or High severity.

The artifact was also improved by adding suspicious IP ranking. This feature groups logs by IP address, calculates total threat score for each IP, counts related events, sorts the results, and displays the most suspicious IP addresses on the dashboard. This enhancement makes the project more realistic because real security monitoring tools usually prioritize activity based on risk level and repeated suspicious behavior.

Course Outcomes and Enhancement Goals

This enhancement supports the course outcome related to designing and evaluating computing solutions using algorithmic principles and computer science practices. The project now shows a much stronger use of algorithms by moving beyond fixed severity rules and applying a more flexible threat scoring system.

This enhancement also supports the outcome related to using well founded and innovative techniques, skills, and tools to implement computing solutions that deliver value and accomplish industry specific goals. In this case, the industry-specific goal is cybersecurity monitoring. The scoring algorithm provides more meaningful alert prioritization and helps simulate the type of risk based analysis used in SOC and SIEM environments.

The enhancement also supports the security mindset outcome because the application now evaluates security events more thoughtfully. Instead of treating each event in isolation, the system considers repeated failed logins and repeated activity from the same IP address. This reflects a more realistic approach to identifying suspicious behavior and potential adversarial activity.

Reflection on the Enhancement Process

The process of enhancing this artifact really helped me to better understand how algorithms can be applied to real world cybersecurity problems. Before this enhancement, the application used simple conditional logic to assign severity. That approach worked for a basic version of the project, but it did not really provide much flexibility or context. By adding weighted threat scoring, I learned how to design a more scalable and meaningful risk evaluation process.

One challenge I faced was deciding how to assign weights to different security events. I had to think through which events should create more risk and how repeated events should affect the final score. This helped me understand the trade offs involved in designing detection logic. If the threshold is too low, the system may create too many high-severity alerts. If the threshold is too high, important suspicious activity may be missed.

I also learned more about how grouping, sorting, and ranking can be used to support security analysis. The suspicious IP ranking feature helped me see how data can be organized to help analysts identify the most important activity first. Overall, this enhancement strengthened my understanding of algorithms, event processing, and alert prioritization in a cybersecurity context.